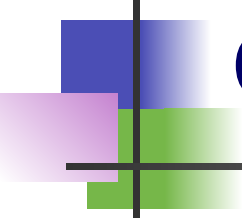




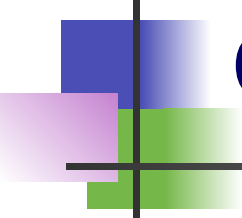
# **Programming Fundamentals**



# Course Structure

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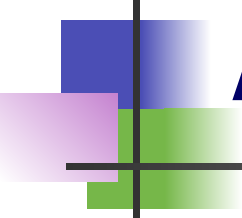
- Lectures
  - About 30 lectures
- Labs
  - About 13
- Tutorial/Problem Solving Sessions
  - About 10
- Assignments
  - Written and Programming Assignments
- Projects (2 Projects)



# Course Grading

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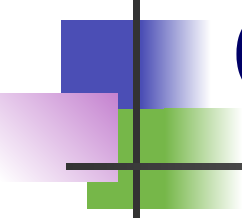
- Assignments (20)
- Quizzes(5)
- Projects(10)
- Midterms(25)
- Final(40)
  
- Grading is ABSOLUTE



# Administrative

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- Attendance policy
  - Minimum acceptable 80% of lectures/sessions.
- Assignments
  - Late submission policy
    - 24 hours (10% Penalty)
    - 48 hours (50% Penalty)
    - >48 hours (100% Penalty)

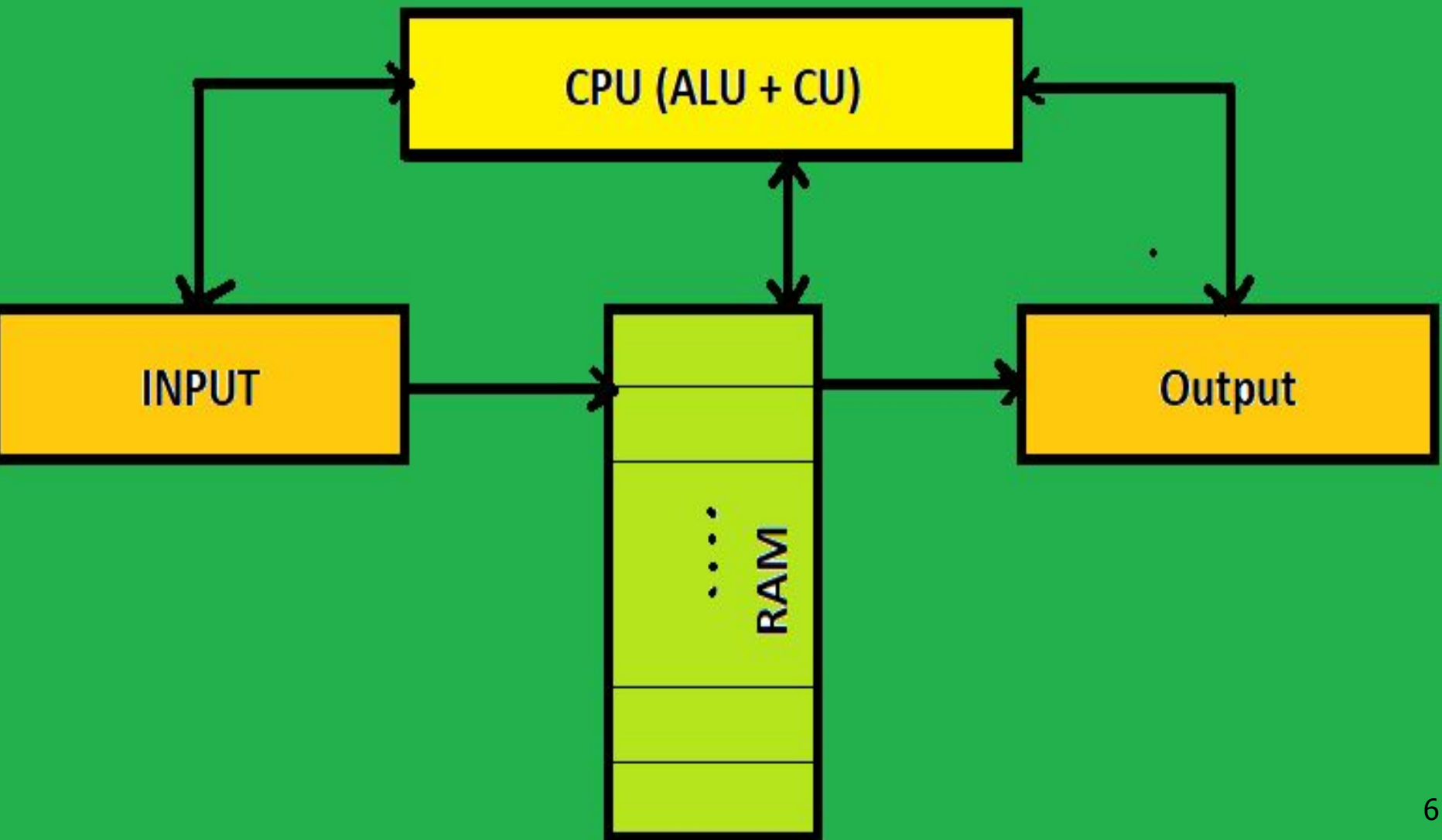


# Course prerequisites

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- A student who passed grade 10 and is willing to work can easily learn all the concepts in this course
- A student might not have seen a computer before this course

# Computer



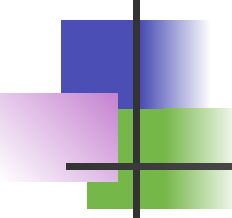


# ALGORITHM 2.0

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## A Language for Writing Instructions

- The name Algorithm is a short for Abu Jafar Muhammad ibn Musa al-Khoarzami
- Which means
  - Abu Jafar (Father of Jafar)
  - Muhammad
  - Ibn Musa (Son of Musa)
  - Al-Khoarzami (Native of Khoarizm)



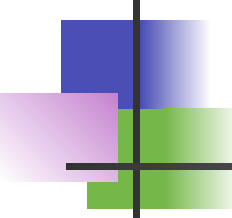
# ALGORITHM 2.0

---

## A Language for Writing Instructions

- **CR** {Create and Name Locations}
  - **CR** Comma separated List of Location\_Names
- **Assignment**
  - Location\_Name = Expression





# ALGORITHM 2.0

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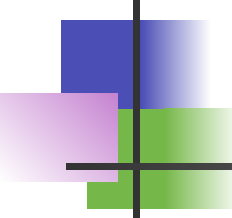
## A Language for Writing Instructions

- **INPUT**

- **IN** comma\_ separated list of location\_names

- **OUTPUT**

- **OUT** comma\_ separated list of expressions



# ALGORITHM 2.0

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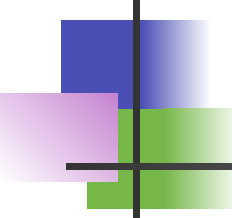
## A Language for Writing Instructions

- **Arithmetic Operations**

- **$+$ ,  $-$ ,  $*$ ,  $/$ , DIV, MOD**

- **Comparison Operation**

- **$<$ ,  $<=$ ,  $>$ ,  $>=$ ,  $==$ ,  $!=$**



# ALGORITHM 2.0

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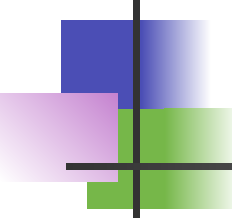
A Language for Writing Instructions

- **Conditional Execution**

**IF** (Expression) **THEN**

INSTRUCTION SEQUENCE

**ENDIF**



# ALGORITHM 2.0

## A Language for Writing Instructions

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- **Conditional Execution**

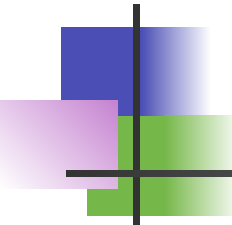
**IF** (Expression) **THEN**

INSTRUCTION SEQUENCE ONE

**ELSE**

INSTRUCTION SEQUENCE TWO

**ENDIF**



# ALGORITHM 2.0

## A Language for Writing Instructions

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- **Conditional Execution**

**IF** (Expression) **THEN**

INSTRUCTION SEQUENCE ONE

**ELSEIF** (Expression) **THEN**

INSTRUCTION SEQUENCE TWO

...

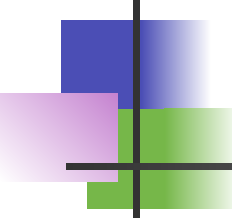
**ELSEIF** (Expression) **THEN**

INSTRUCTION SEQUENCE

**ELSE**

INSTRUCTION SEQUENCE

**ENDIF**



# ALGORITHM 2.0

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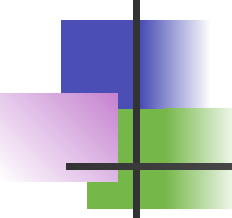
A Language for Writing Instructions

- **Repeated Execution**

**REPEAT** (expression) **TIMES**

INSTRUCTION SEQUENCE

**END REPEAT**



# ALGORITHM 2.0

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A Language for Writing Instructions

- **Repeated Execution**

**REPEAT WHILE** (expression)

INSTRUCTION SEQUENCE

**END REPEAT**